



Define, foster, and assess student and teacher AI literacy and competency for all: Current status and future research direction

1. Introduction

AI technologies are expected to become more extensively incorporated into various facets of our daily lives, including education, entertainment, communication, and healthcare. They change the ways we live, learn, and work. Teaching AI topics in P-K16 and adulthood is an important global strategic initiative in educating the next generation with good AI literacy skills [6,3, 7], known as AI for all. Individuals should possess a fundamental knowledge of AI, known as AI literacy. AI literacy refers to a set of competencies that enables individuals to use and evaluate AI applications and to communicate and collaborate effectively with them, as defined by Long et al. [12]. People with strong AI literacy understand fundamental AI concepts such as big data, machine learning, computer vision, computer voice, and natural language processing, as well as the ethical and legal considerations such as AI bias and ethical principles [7,17]. They are also aware of the potential risks and rewards associated with AI. This literacy is seen as one of the essential competencies for the future workforce [3,20]. Moreover, the emergence of generative artificial intelligence (GenAI) has had a disruptive effect on education [5]. GenAI systems, like ChatGPT, are capable of generating text, graphics, and other digital material for students to engage with. Students may use them to enhance their learning or finish assignment without understanding. GenAI's impact on learning can vary between positive and negative outcomes [4]. Using GenAI requires more than just understanding AI; students need to demonstrate responsibility, ethics, confidence, and proficiency to fully leverage the benefits of GenAI in their educational pursuits, which is considered AI competency [3]. AI literacy and competency are crucial for both students and teachers in education. Many questions remain in every aspect of this initiative - AI for All. This special issue sought to address these problems by enhancing AI literacy and competency developments.

2. Overview of the articles

This special issue consists of 18 articles authored by scholars from various regions: Canada, Finland, Germany, Hong Kong, Iran, Saudi Arabia, South Africa, Sweden, and the United States of America. They cover more of the important research areas that promote AI for all. They discussed the following five areas: (i) definition, (ii) instructional design and assessment, (iii) teacher education, (iv) educational policy, and (v) AI in education.

a. Definitions and assessment of AI literacy and competency.

Research on AI literacy and competency is still in its early stages,

with the majority of the work undertaken in the previous few years. More studies are needed to develop a robust, comprehensive, and precise understanding of what non-professionals should learn. The articles in the special issues present some definitions of literacy and competency and their relationship with assessment. Chiu and colleagues [4] discussed the differences between AI literacy and AI competency by presenting their definitions. AI literacy is often knowledge- and skills-focused. AI competency goes beyond AI competency and includes attitude, beliefs, and self-reflective mind-sets. Their definition of AI is "AI competency," defined as "an individual's confidence and ability to clearly explain how AI technologies work and impact society, as well as to use them in an ethical and responsible manner and to effectively communicate and collaborate with them in any setting. They should have the confidence and ability to self-reflect on their AI understanding for further learning. It focuses on how well individuals use AI in beneficial ways." Furthermore, they recommended moving from AI literacy to AI competency for teachers as well as students. Both teachers and students can use AI in a safe, ethical, responsible, and productive way. Almatrafi and colleagues [1] used a systematic review to define AI literacy by suggesting a framework comprising six key constructs of AI literacy: Recognize, Know and Understand, Use and Apply, Evaluate, Create, and Navigate Ethically. Knoth and colleagues [9] propose a 2×2 framework to define AI literacy. The framework uses three aspects—knowledge/cognition, skill/behavior, value/attitude—to match generic and domain-specific aspects (medicine, engineering, and education) and AI ethics literacy. Stolpe and Hallström [19] suggested AI should be viewed as multiple literacies from the perspectives of technology education. They also proposed a framework comprising three major components: Technological scientific knowledge, Technical skills Socio-ethical technical, understanding. Overall, these definitions offered new perspectives on the existing definitions and laid the groundwork for research on instrument development. Some of the articles presented items that could be used to measure AI literacy.

b. **Instructional designs and assessment for AI literacy and competency.** The articles in this issue explored fostering student AI literacy as the second area. For example, Markus and colleagues [13] designed and implemented online training courses on intelligent voice assistants to examine learners' AI literacy development. They used learning sciences and media pedagogy perspectives to interpret their findings, demonstrating that self-determined interaction with voice-based AI systems reduced misunderstanding, enhanced AI literacy, and sparked curiosity in studying AI. Korte and colleagues

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[10] used flipped learning to enhance students' AI and media literacy, while also pinpointing certain design features. Parker and colleagues (2024) conducted a study on the influence of ChatGPT on grading [16]. They proposed that AI-generated assessments can provide high-quality assignments and advocated for reforms in assessment processes. These findings suggest that instructors and teaching assistants require training in AI, specifically in teacher education, see the next area.

- c. **Teacher education.** Teachers and institutions impact the development of student AI literacy and competency, as explored in this and the following areas. This area underscores the importance of teacher education in promoting AI for all. Sperling and colleagues (2024) proposed that AI literacy is a growing research focus in education worldwide but is often overlooked in the realm of teacher education. The research does not comprehensively encompass teachers' professional expertise. Law [11] emphasized the importance of continuous professional growth for teachers and teacher educators in order to facilitate informed decision-making and the successful integration of GenAI technologies. Another significant issue in this area is "how to design an effective teacher education program." Ayanwale and colleagues (2024) investigated many aspects of AI literacy and discovered that AI understanding correlates with AI use, detection, ethics, creation, and problem-solving. There is no connection between AI knowledge and emotion regulation. When designing teacher education activities, they acknowledge a trade-off between the use and development of AI tools, emphasizing the growth of ethical knowledge associated with attitudes towards AI use. Ding and colleges [8] investigated how a case-based AI professional development program impacted teachers' AI literacy growth. To successfully enhance teacher AI literacy, it is critical to integrate direct instruction with case-based discussions in AI-related professional development programs. AI literacy is hindered by the intricacy of the case problems. Overall, the articles suggested we need more studies on how to design effective teacher education.
- d. **Educational policies for AI.** Instructional policy influences the development of AI literacy in students and teachers. Only one article in this special issue directly addresses this area. Moorhouse and colleagues (2024) examined the policies of top universities and recommended that institutions embrace the use of GenAI and develop assessment guidelines to aid teachers in its implementation [15]. They suggest that including GAI in the assessment process might be advantageous due to its relevance in today's educational and professional environment. Teachers will need to acquire a new skill set called GenAI assessment literacy. Overall, Without policy changes, the sustainability of changes in learning and working environments will not be maintained.
- e. **AI in education.** While the main focus of this issue is on the AI literacy and competency, several articles also explored the influence of AI on learning. This highlights the significance of AI education, as students and educators should possess the AI understanding to evaluate the effectiveness of AI-based learning platforms. Memarian and Doleck [14] conducted research on reinforcement learning in education, which has gained significant interest for providing feedback to students. Law [11] examined the impact of GenAI on education, supporting the incorporation of GenAI technology in instructional practices but acknowledging potential risks such as ethical issues. These articles recommended that both teachers and students improve their AI literacy and competency to successfully use AI in teaching and learning by understanding its capabilities and limitations.

In sum, the articles advocates AI literacy and competency for all, and sees AI education as foundational learning. Their findings outlined the definitions of AI literacy and competency, proposed strategies for designing student and teacher activities related to AI, advocated for the formation of educational policies on AI, and emphasized that AI literacy

and competency are essential for using AI in education.

3. Future research direction

AI literacy and competency for all is a growing field of research. In this special issue, the articles propose 10 potential research directions aimed at stimulating discussions among scholars and practitioners on reimagining AI education and influencing the future of learning and teaching.

- a. **Assessment of AI literacy and competency.** The articles explored the definitions of AI literacy and competency and proposed the necessity for more instruments [1,3]. The definitions in this issue might establish a foundation for this future course. We recommend using the definitions to develop and validate new instruments in other contexts and perspectives.
- b. **New AI-related literacies and competencies.** Some of the articles discussed the concepts of multiliteracies for AI, as AI is an interdisciplinary field [3,9]. AI literacy interconnects with digital, data, and media literacy. Future studies should explore a novel literacy framework for both students and teachers.
- c. **Upscaling and sustaining AI teacher education.** Most current research has focused on one-off professional development programs aimed at enhancing teachers' AI literacy and competency [2,18]. This might not be efficient. Research is required to upscale and sustain teacher education. Future studies should explore approaches to improving teachers' beliefs about learning AI and using AI in their teaching, as well as creating and sustaining learning communities.
- d. **Evaluation of the AI literacy program.** AI is new to non-engineering. Most studies suggest what learning objective (definition of AI literacy) and learning content should be included in the AI learning program. They disregard the program's standards. We suggest future studies should recommend establishing a specific set of criteria or framework for assessing the program.
- e. **Human-AI interaction.** Some publications have explored how learners engage with AI, but their findings are limited [4]. Future studies should investigate the impact of AI literacy and competency on interactions between humans and AI, as well as the specific types of interactions required by students and teachers.
- f. **Equity and accessibility.** There is ongoing debate on whether AI might widen the digital divide [20]. We included this topic in the special issue call, but no accepted articles have emerged. It could be due to a lack of empirical research. Future research might explore the impact of social-economic status and geography (rural areas) on the development of AI literacy and competency. We recommend conducting comparative studies across different regions.
- g. **Inclusive and divisive AI education.** AI literacy is considered crucial for future education and job opportunities. It is essential to ensure that everyone learns AI efficiently (Xia et al., 2024). Despite our inclusion of this topic in the special call, we have not received any accepted papers. Hence, we recommend that future research focus on designing AI education to meet the needs of learners with diverse characteristics, including gender, minority groups, digital literacy, and media literacy.
- h. **Educational policy.** AI policy development for higher education has more studies ([5]; Moorhouse et al., 2024), compared to K-12 education. Future research is required to reexamine studies on how AI influences educational policy, to help redefine the vision of future AI education in K-12.
- i. **Ethics of AI.** Like in this special issue, where the ethical dimension of AI is not very pronounced, limited literature infuses ethics orientation into AI learning materials [19]. In addition, didactics of ethics are not examined, such as discussions of typical or important teaching methods (such as dilemma questions). Future research should consider the idea of an ethics-infused teaching approach,

including the ethical implications of AI applications that are becoming ubiquitous.

- j. **AI integration into core subjects.** We can leverage AI across the curriculum to democratize AI for all learners. While research has recently begun on how to integrate AI into subject domains (Korte et al., 2024), more work is required to develop and co-design activities that support the integration of AI into STEM and non-STEM subjects. Future research should also consider how data science and working with data support students learning of AI.

Overall, AI has a significant impact on the world's education, making it essential for instructors and students to possess good literacy and competency to effectively interact with AI in their lives, learning, and jobs. We need to rethink several aspects of education in the AI age. We believe this special issue will enhance our discussions and research on the future of learning and education in relation to AI.

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